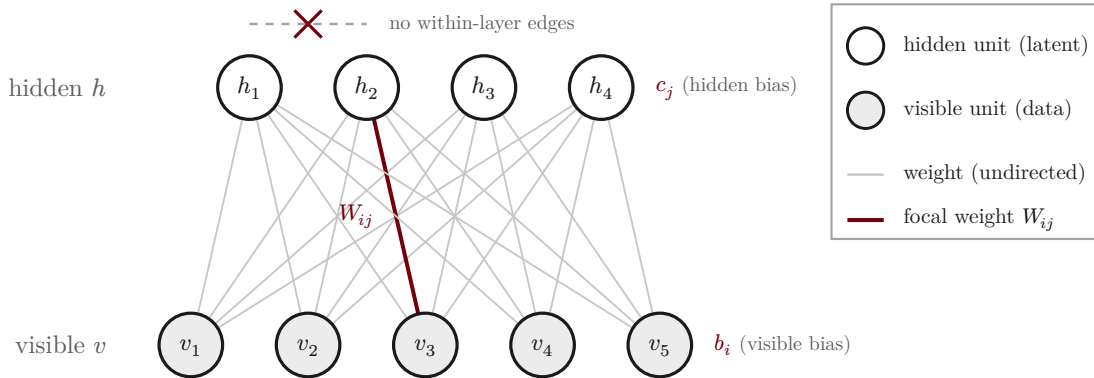


Restricted Boltzmann Machine

bipartite undirected graph: every visible–hidden pair coupled by a symmetric weight W_{ij}



$$E(v, h) = -b^\top v - c^\top h - v^\top W h, \quad p(v, h) = e^{-E(v, h)} / Z$$

restriction $\Rightarrow$ factorial conditionals: $p(h \mid v) = \prod_j p(h_j \mid v)$, $p(v \mid h) = \prod_i p(v_i \mid h)$